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Design and Implementation of a 4DOF Robotic Arm Using Arduino and Servo Motors

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Report Writing Contribution

Project title: Design and Implementation of a 4DOF Robotic Arm Using Arduino and Servo Motors

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1.2 Problem Definition	Yousef
1.3 Project Aim and Objectives	Yousef
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Abstract

This project presents the design and implementation of a 4-degree-of-freedom (4DOF) robotic arm using an Arduino Uno and a PCA9685 servo driver. The robotic arm was developed as an application of rigid body dynamics principles, rotational motion, and embedded control systems. The system consists of four servo motors responsible for controlling the base, shoulder, elbow, and gripper joints, allowing the arm to perform coordinated movements and simple object-manipulation tasks.

The system was powered using two 3.7V lithium-ion batteries connected in series. A DC step-down buck converter was used to regulate the output voltage to 5V, providing stable power for the electronic components and servo motors. The PCA9685 servo driver generated stable PWM signals while reducing the processing load on the microcontroller. The control algorithm was programmed using the Arduino IDE and included smooth motion interpolation to reduce sudden movements and servo jitter during operation.

The project demonstrates important engineering concepts including degrees of freedom, torque generation, angular motion, and kinematic control of linked rigid bodies. Experimental testing showed successful operation of the 4DOF robotic arm with stable multi-joint movement and accurate positional control. This project enhanced understanding of robotics, rigid body dynamics, mechatronics, and microcontroller-based automation systems while providing practical experience in both mechanical and electronic system integration.

Chapter I: Introduction

1.1 Background of the Study

Robotic arms are among the most important applications of modern robotics and automation systems. They are widely used in industries such as manufacturing, automotive assembly, medical engineering, packaging, and material handling due to their ability to perform repetitive tasks with high precision and efficiency. A robotic arm is designed to imitate the motion of a human arm through a series of connected rigid links and rotating joints that provide controlled movement in different directions.

In engineering, robotic arms are closely related to Rigid Body Dynamics because their motion depends on the analysis of forces, torques, angular displacement, velocity, and acceleration of interconnected rigid bodies. Each joint of the robotic arm contributes a specific degree of freedom (DOF), allowing the end effector to reach different positions and orientations within the workspace.

The development of low-cost microcontrollers and embedded systems has made robotic arm design more accessible for educational and research purposes. Platforms such as the Arduino Uno provide simple and efficient control of actuators and sensors, making them suitable for robotic applications. In addition, the PCA9685 enables stable control of multiple servo motors using PWM signals while reducing the computational load on the microcontroller.

Servo motors are commonly used in robotic arms because they allow accurate angular position control and compact mechanical design. By coordinating the motion of multiple servos, robotic systems can perform tasks such as lifting, rotating, gripping, and object manipulation. Proper power management is also essential in robotic systems; therefore, a DC step-down buck converter was used in this project to regulate the voltage from two 3.7V lithium-ion batteries to a stable 5V supply suitable for the electronic components and motors.

This project focuses on the design and implementation of a 4DOF robotic arm as a practical application of rigid body dynamics, mechatronics, and embedded control systems. The project combines mechanical structure design, electronic circuit integration, and motion programming to achieve smooth and coordinated robotic arm movement.

1.2 Problem Definition

In modern industrial environments, many manufacturing and material-handling operations require repetitive movement of objects from one location to another. Performing these tasks manually can lead to reduced efficiency, increased human effort, and a higher chance of positioning errors, especially in continuous production processes. As industries continue to move toward automation, robotic arms have become an effective solution for improving accuracy, speed, and operational reliability.

The main problem addressed in this project is the development of an autonomous robotic arm capable of performing a basic pick-and-place operation. The robotic system is designed to pick up an object, represented in this project by a small box, and transfer it to another specified location with controlled and coordinated movement.

Designing such a system involves several engineering challenges related to rigid body dynamics, including the control of rotational motion, synchronization of multiple joints, torque generation, and

maintaining stable movement during operation. In addition, the robotic arm must provide sufficient degrees of freedom to allow proper positioning of the end effector while ensuring smooth and accurate motion.

To address these challenges, this project implements a 4DOF robotic arm controlled using an Arduino Uno and a PCA9685. The system uses servo motors to achieve precise angular control of each joint, enabling the arm to successfully perform object-transfer tasks in a simple industrial-simulation environment.

1.3 Project Aim and Objectives

Project Aim

- To design and implement a 4-degree-of-freedom robotic arm capable of performing autonomous pick-and-place operations using embedded control systems and rigid body dynamics principles.

Project Objectives

- To design and construct the mechanical structure of the robotic arm.
- To control four servo motors independently for coordinated joint movement.
- To use an Arduino Uno for robotic arm control and programming.
- To use a PCA9685 to generate stable PWM signals for the servo motors.
- To apply concepts of rigid body dynamics in rotational motion and robotic movement.
- To implement smooth motion control to reduce servo jitter and sudden movements.
- To develop a stable power supply system using lithium-ion batteries and a DC step-down buck converter.
- To demonstrate the robotic arm's ability to move a box from one specified location to another, accurately and reliably.
- To gain practical experience in robotics, embedded systems, and mechatronic system integration.

1.4 Proposed Solution

The proposed solution for this project is the development of a compact and low-cost 4DOF robotic arm capable of performing autonomous pick-and-place operations. The system is designed to simulate a simple industrial robotic arm that can pick up a box from one location and place it at another specified location with controlled and coordinated movement.

The robotic arm consists of four servo motors that provide rotational motion for the base, shoulder, elbow, and gripper joints. These joints work together to provide the required degrees of freedom for object manipulation and positioning. The system is controlled using an Arduino Uno, while a PCA9685 is used to generate stable PWM signals and improve servo control performance.

To ensure stable power delivery, the system uses two 3.7V lithium-ion batteries connected in series along with a DC step-down buck converter to regulate the voltage to 5V, suitable for the electronic components and servo motors. The robotic arm is programmed using the Arduino IDE, where motion

algorithms are implemented to achieve smooth movement, reduce servo jitter, and improve positional accuracy.

The proposed system combines mechanical design, electronic control, and rigid body dynamics concepts to create a functional robotic arm capable of demonstrating practical robotic manipulation and coordinated multi-joint motion.

1.5 Report Organization

This report is organized into six main chapters, each focusing on a specific part of the design, development, and evaluation of the 4DOF robotic arm project. The structure is as follows:

- Abstract — provides a brief overview of the project, including its purpose, methodology, and key outcomes.
- Acknowledgement — expresses gratitude to individuals who contributed to the completion of the project.
- Chapter I: Introduction — introduces the project, including the background of the study, problem definition, project aim and objectives, proposed solution, and overall report organization.
- Chapter II: Hardware Components — describes all the electronic and mechanical components used in the system, including the Arduino Uno, the PCA9685, servo motors, and the power supply system, and includes a cost analysis of the robotic arm components.
- Chapter III: Mechanical Design — explains the mechanical structure of the robotic arm, including the design of links, joints, degrees of freedom, and overall assembly.
- Chapter IV: Circuit Design — presents the electrical circuit design, wiring connections, and power distribution system used to control the robotic arm.
- Chapter V: Results and Discussion — discusses the experimental results, system performance, motion accuracy, and analysis of the robotic arm operation.
- Chapter VI: Conclusion and Future Work — summarizes the findings of the project and suggests possible improvements and future enhancements, such as advanced control methods and automation features.
- References — lists all sources, datasheets, and references used throughout the project development.

Chapter II: Hardware Components

2.1 Preface

This chapter presents the hardware components used in the development of the 4DOF robotic arm and explains their main functions within the system. The design is based on a microcontroller-based architecture using an Arduino Uno as the main controller and a PCA9685 for generating stable PWM signals to control multiple servo motors.

The system also includes servo motors, a mechanical structure, and a power supply system using lithium-ion batteries with a DC step-down buck converter to provide a stable 5V output. These components were selected to ensure a low-cost, reliable, and functional robotic arm suitable for demonstrating basic robotic motion and rigid body dynamics principles.

2.2 Components Description

In this section, each component used in the project is described in detail, including its specifications and function.

1. PCA9685 — Motor Driver

This motor driver is important for controlling the servos that actuate the robotic arm. It uses the I2C protocol to communicate with the Arduino Uno microcontroller, allowing control of the servo motors.

2. SG90 Micro Servo Motors

These small, lightweight servo motors are used to provide the necessary movement to the robotic arm. Each motor has a limited range of motion, but their combination enables the arm to achieve four degrees of freedom (DOF).

3. Arduino Uno

The Arduino Uno works as the brain of the robotic arm, controlling the motors and processing inputs. It is the microcontroller that enables communication between the motors and the other components in the system.

4. Battery Holder and Li-ion Batteries

The battery holder and two 3.7V Li-ion batteries supply power to the robotic arm system.

5. Robot Arm 3D-Printed Model

The 3D-printed robotic arm is the physical structure of the system. It consists of a series of links and joints that allow for four axes of movement. This arm is built to interact with its environment by grasping, moving, and positioning objects.

6. LM2596 — Step-Down Converter

This step-down voltage regulator ensures that the power supplied to the microcontroller and other components is stable and within the required voltage range, preventing damage from overvoltage.

7. Jumper Wires

Jumper wires are used to establish connections between the various components, such as between the Arduino Uno and the motor driver, or between the motor driver and the servo motors.

2.3 Cost Analysis of the Robotic Arm Components

This section presents the cost of each hardware component used in the robotic arm project. The total cost is calculated by adding the price of each component, as listed below.

Component	Quantity	Unit Price (LE)	Total Price (LE)
PCA9685 — Motor Driver	1	210	210
SG90 Micro Servo Motors	5	85	425
Arduino Uno — Microcontroller	1	450	450
Battery Holder	1	40	40
Li-ion Batteries (3.7V)	2	85	170
3D-Printed Robot Arm Body	1	700	700
LM2596 — Step-Down Converter	1	45	45
Jumper Wires	1 set	35	35
Total Cost			2,075 LE

Summary of Costs

- Total hardware cost: 2,075 LE.
- The most expensive single part, at 700 LE, is the 3D-printed body.
- The motor driver and microcontroller contribute significantly to the total cost, while components such as the jumper wires and battery holder are more affordable.

Strengthening the Project

- Add additional sensors for better accuracy.
- Use higher-quality servos to improve performance, especially in complex tasks.
- Increase the robot's degrees of freedom for more flexibility.
- Add wireless control (Bluetooth / Wi-Fi) for remote operation.

Losses

One servo motor was lost during assembly, affecting precision and increasing costs for replacement.

Chapter III: Mechanical Design

3.1 Preface

This chapter describes the mechanical design of the 4DOF robotic arm, focusing on its structure, joints, and overall assembly. The design is based on a system of interconnected rigid links that replicate human arm motion through rotational joints, allowing controlled movement in multiple directions.

The mechanical configuration is developed to provide four degrees of freedom, enabling the robotic arm to perform basic pick-and-place operations. The design is directly related to concepts in rigid body dynamics, where motion, torque, and link interactions are analysed to ensure stable and efficient operation of the system.

3.2 Mechanical Design Description

The robotic arm is designed as a 4-degree-of-freedom structure consisting of rigid links connected through rotational joints. Each joint is actuated using a servo motor, allowing controlled angular motion that collectively defines the position and orientation of the end effector.

The main structure includes a base joint for horizontal rotation, a shoulder joint for vertical lifting, an elbow joint for extending and retracting the arm's reach, and a gripper mechanism for holding objects. These joints are arranged to simulate the motion of a human arm while maintaining mechanical stability and balance.

The links are designed to be lightweight yet strong enough to support the load of small objects such as a box. The geometry of each link is selected to ensure sufficient workspace coverage while minimizing excessive torque requirements on the servo motors. This helps improve motion efficiency and reduces mechanical stress during operation.

The overall design is based on principles of rigid body dynamics, where each link is treated as a rigid body and each joint contributes to the system's degrees of freedom. The interaction between torque, angular displacement, and load distribution is considered to ensure smooth and stable movement of the robotic arm during pick-and-place tasks.

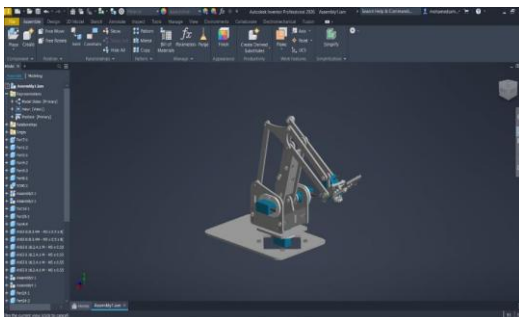


Figure 3.1 — CAD assembly of the robotic arm (Autodesk Inventor) and rendered 3D model

3.3 Control Code

The robotic arm is programmed using the Arduino IDE. The PCA9685 servo driver library is used to generate PWM signals for each joint, and a smooth-motion function interpolates between angles to reduce servo jitter and sudden movements during the pick-and-place sequence.

```
#include <Adafruit_PWMServoDriver.h>
#include <Wire.h>

Adafruit_PWMServoDriver pwm = Adafruit_PWMServoDriver();

#define SERVOMIN 150
#define SERVOMAX 600

int angleToPulse(int angle) { return map(angle, 0, 180, SERVOMIN, SERVOMAX); }

// positions
#define MAIN_HOME 0
#define MAIN_PICK 60

#define ARM_UP 45
#define ARM_DOWN 90

#define CLAMP_CLOSE 0
#define CLAMP_OPEN 90

void moveServoSmooth(int channel, int startAngle, int endAngle, int stepDelay) {
  if (startAngle < endAngle) {
    for (int a = startAngle; a <= endAngle; a++) {
      pwm.setPWM(channel, 0, angleToPulse(a));
      delay(stepDelay);
    }
  } else {
    for (int a = startAngle; a >= endAngle; a--) {
      pwm.setPWM(channel, 0, angleToPulse(a));
      delay(stepDelay);
    }
  }
}

void setup() {
  pwm.begin();
  pwm.setPWMFreq(50);
  delay(1000);

  // initial state
  pwm.setPWM(0, 0, angleToPulse(MAIN_HOME));
  pwm.setPWM(1, 0, angleToPulse(ARM_UP));
  pwm.setPWM(2, 0, angleToPulse(ARM_UP));
  pwm.setPWM(3, 0, angleToPulse(CLAMP_CLOSE));

  delay(2000);
}

void loop() {
  // 1. rotate to object (smooth)
  moveServoSmooth(0, MAIN_HOME, MAIN_PICK, 15);

  // 2. clamp open
  moveServoSmooth(3, CLAMP_CLOSE, CLAMP_OPEN, 10);

  // 3. arms go down (smooth both)
  moveServoSmooth(1, ARM_UP, ARM_DOWN, 15);
  moveServoSmooth(2, ARM_UP, ARM_DOWN, 15);

  // 4. clamp close
  moveServoSmooth(3, CLAMP_OPEN, CLAMP_CLOSE, 10);

  // 5. lift arms
  moveServoSmooth(1, ARM_DOWN, ARM_UP, 15);
  moveServoSmooth(2, ARM_DOWN, ARM_UP, 15);

  // 6. return home
  moveServoSmooth(0, MAIN_PICK, MAIN_HOME, 15);

  // 7. open clamp
  moveServoSmooth(3, CLAMP_CLOSE, CLAMP_OPEN, 10);
}
```

Figure 3.2 — Arduino control code for smooth multi-joint servo motion

Chapter IV: Circuit Design

4.1 Preface

This chapter presents the circuit design of the 4DOF robotic arm, focusing on how the electronic components are connected and integrated to achieve reliable control and power distribution. It explains the wiring configuration between the controller, servo driver, actuators, and power supply system.

The circuit is built around an Arduino Uno and a PCA9685, which work together to generate and manage PWM signals for precise servo motor control. The chapter also describes the power management system, including the use of lithium-ion batteries and a DC step-down buck converter to ensure stable voltage delivery to all components.

4.2 Circuit Description

4.2.1 Microcontroller Unit

The system is controlled using an Arduino Uno, which acts as the main processing unit. It sends control commands and communicates with the servo driver using the I2C protocol (SDA and SCL pins).

4.2.2 Servo Driver Module

A PCA9685 is used to generate stable PWM signals for all servo motors. It receives control signals from the Arduino and distributes independent PWM outputs to each servo channel, allowing precise multi-joint control without overloading the microcontroller.

4.2.3 Servo Motors

Four servo motors are connected to the PCA9685 output channels. Each servo is responsible for one joint of the robotic arm (base, shoulder, elbow, and gripper). The servos receive both PWM control signals and external power to ensure reliable torque and movement.

4.2.4 Power Supply System

The power system consists of two 3.7V lithium-ion batteries connected in series. The combined voltage is regulated using a DC step-down buck converter to provide a stable 5V output. This regulated supply powers the servo motors and the PCA9685 module, ensuring consistent performance.

4.2.5 Grounding and Connections

A common ground is shared between the Arduino, PCA9685, servo motors, and power supply system. This ensures proper signal reference, stable communication, and prevents control errors during operation.

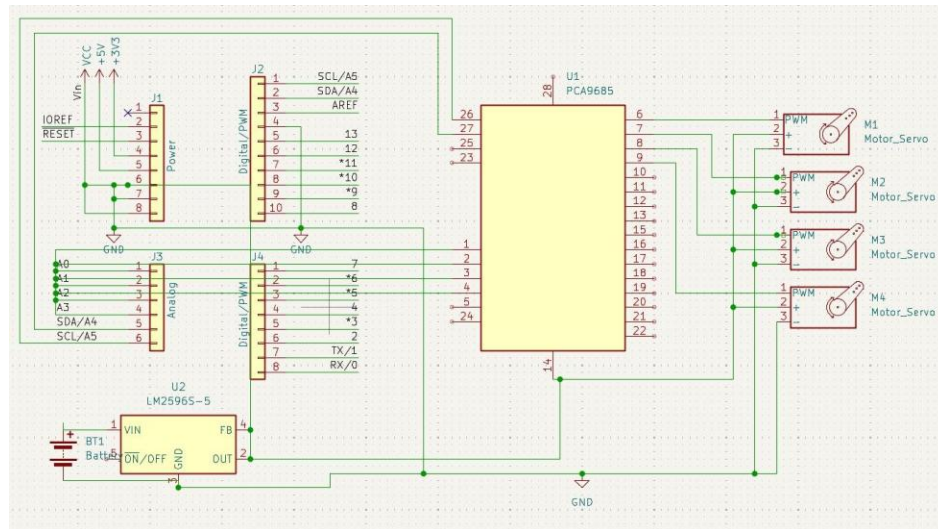


Figure 4.1 — Full circuit schematic: Arduino, PCA9685, servo motors, and power supply

Chapter V: Results and Discussion

5.1 System Testing

After completing the assembly of the robotic arm, the system was tested to check that all parts were working correctly. The Arduino Uno was connected to a computer and the control code was uploaded. The PCA9685 motor driver was connected to the Arduino using the I2C communication pins (SDA and SCL). The servo motors were connected to the motor driver outputs, and the power supply was connected through the LM2596 step-down converter.

When the system was powered on, the robotic arm moved according to the programmed angles. The base servo rotated the arm left and right. The shoulder and elbow servos moved the arm up and down. The gripper servo opened and closed to hold small objects such as a plastic bottle cap and a small box.

The arm was tested several times to confirm that the movement was smooth and repeatable. The servo motors responded to the signals without noticeable delay. The PCA9685 driver handled all four servo motors without issue, and the Arduino did not overheat or reset during operation.

5.2 Observations

During testing, several important points were noted:

- The arm could lift light objects easily but struggled with heavier items, since the SG90 servos are small and do not provide high torque.
- The gripper sometimes needed adjustment to hold objects firmly; the gripper design could be improved for better grasping.
- The battery voltage dropped slightly after roughly 30 minutes of continuous use. The LM2596 kept the output voltage stable, but the batteries needed recharging after extended use.
- One servo motor was damaged during assembly when its horn was forced into place. It was replaced with a new unit — a reminder to be more careful when assembling mechanical parts.

5.3 Discussion

The results showed that the project achieved its main goals. The robotic arm moved with four degrees of freedom and could pick and place objects successfully. Using the PCA9685 motor driver solved the problem of limited PWM pins on the Arduino Uno, and I2C communication worked well, allowing multiple servos to be controlled with only two data wires.

The power supply design was also successful. The two Li-ion batteries provided sufficient voltage, and the LM2596 step-down converter gave a stable 5V output for the servos and Arduino. Without this regulator, the servos could have received excessive voltage and been damaged.

However, there were some limitations. The SG90 servos are inexpensive but not very strong; larger or more powerful servos would allow the arm to lift heavier objects. In addition, the arm is controlled using pre-programmed angles rather than sensors or feedback, meaning it cannot automatically adjust its movement if something blocks its path.

Overall, the system performed as expected for a low-cost educational project. The total cost of 2,075 LE is reasonable for students, and the components are easy to find in local markets.

Chapter VI: Conclusion and Future Work

6.1 Conclusion

This project involved building a 4DOF robotic arm using simple, low-cost parts. An Arduino Uno served as the main controller, a PCA9685 motor driver controlled the servos, SG90 micro servo motors provided movement, and a power system combining Li-ion batteries with an LM2596 step-down converter supplied stable power.

The robotic arm was successfully assembled and tested. It moved with four degrees of freedom and could pick up and place small objects. The motor driver made it straightforward to control four servos, and the power supply kept the system running safely. The process involved connecting electronic components, writing Arduino code, and solving problems that arose during assembly.

The project helped build an understanding of the basics of robotics, control systems, and embedded systems. Challenges such as losing one servo motor during assembly and finding the right voltage levels were resolved by working together and consulting the course instructor when needed.

In the end, the team built a working robotic arm that meets the project requirements. It is not perfect, but it provides a solid starting point for learning about robotics.

6.2 Future Work

If this project is continued in the future, several improvements could be made:

- Stronger servo motors: MG996R or DS3218 servos could replace the SG90 units, offering higher torque and the ability to lift heavier objects.
- Closed-loop control: position-feedback sensors could be added to each joint, allowing the arm to know its exact position and correct errors during movement.
- Wireless control: a Bluetooth or Wi-Fi module could allow the arm to be controlled from a mobile phone or computer without a USB cable.
- More degrees of freedom: a wrist-rotation joint or a sliding base could be added to give the arm greater flexibility and reach.
- Better gripper design: the gripper could be redesigned with softer material or a different shape to hold various objects more securely.
- Computer vision: a camera could be added to allow the system to detect and locate objects automatically.

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